OpenHIE Auth & Bring-up Setup (Windows)

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Local / single-host bring-up on Windows 10/11. PowerShell for everyday commands; the `keycloak/*.sh` scripts are Bash — run those from **Git Bash** (ships with Git for Windows) or WSL. `<like-this>` = placeholder.

> In PowerShell `curl` is an alias for `Invoke-WebRequest`. Use **`curl.exe`** everywhere below — the real curl, bundled with Windows 10 1803+.

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0. Prerequisites

Install:

- **Docker Desktop** (WSL 2 backend) — https://docs.docker.com/desktop/install/windows-install/ (verify: `docker compose version`)
- **Git for Windows** — https://git-scm.com/download/win (gives you Git Bash)
- **WireGuard for Windows** — https://www.wireguard.com/install/ (only for §3 Option B)
- **Python 3** — Microsoft Store or https://www.python.org/downloads/windows/ (the Store build registers `python3`, which `onboard-hospital.sh` needs)
- **jq** — `winget install jqlang.jq`, then reopen the shell (Git Bash has no `jq`)

The `keycloak/*.sh` scripts need `bash`, `docker`, `openssl`, `curl`, `jq`, `python3` all on PATH — Git Bash provides all but `jq` and `python3`. The repo's `.gitattributes` keeps the scripts LF so they run under Git Bash unmodified; if you see `$'\r': command not found`, your checkout converted them to CRLF — run `git config core.autocrlf false` and re-checkout.

Get the code (PowerShell):

```
git clone https://github.com/ETech-Nepal/openhim.git openhie
cd openhie
'mediator','admin-mediator','patient-portal-mediator','fhir-intake-mediator' |
  ForEach-Object { Copy-Item "$_\.env.example" "$_\.env" }
```

The examples are preconfigured for the compose network. The only value you must edit for bring-up is `OPENHIM_PASSWORD` in all four — done in §2c, once you have the password from §2b.

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1. Containers

```
docker compose up -d
docker compose ps
```

Running: `hapi-db`, `mongo-db`, `openhim-core`, `openhim-console`, `fhir`, `keycloak`, `mediator`, `admin-mediator`, `patient-portal-mediator`, `fhir-intake-mediator`.

> **Never** `docker compose down` (wipes Keycloak's H2). Use `docker compose stop` / `docker compose start`. Mediators crash-loop until steps 2 and 5 are done.

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2. OpenHIM first login

2a. Accept the self-signed cert

1. Open https://localhost:8080/heartbeat in a browser, click through the warning.
2. CLI needs `-k`: `curl.exe -k https://localhost:8080/heartbeat`

2b. Root password

1. Console http://localhost:9000, log in `root@openhim.org` / `openhim-password`.
2. Set a new password. Record it.

2c. Propagate to mediators

Edit `OPENHIM_PASSWORD` in `mediator\.env`, `admin-mediator\.env`, `patient-portal-mediator\.env`, `fhir-intake-mediator\.env` (Notepad or `notepad mediator\.env`), then:

```
docker compose up -d --force-recreate mediator admin-mediator patient-portal-mediator
        fhir-intake-mediator
```

> `restart` does **not** reload `.env` — use `up -d --force-recreate`.

Console → **Mediators**: all four green.

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3. Reach JeMPI

The mediator talks to JeMPI at `JEMPI_BASE_URL` in `mediator\.env`. Pick one. After changing it: `docker compose up -d --force-recreate mediator`.

Option A — JeMPI public IP (recommended on Windows)

```
# mediator\.env
JEMPI_BASE_URL=http://202.51.71.81:50000/JeMPI
```

Check: `curl.exe http://202.51.71.81:50000/JeMPI/countRecords`

No tunnel, no routing fixes. Traffic crosses the public internet unencrypted (JeMPI has no TLS/auth of its own) — test data only; use Option B for real patient data.

Option B — WireGuard tunnel

Keeps the default `JEMPI_BASE_URL=http://10.8.0.1:50000/JeMPI`.

1. Open **WireGuard** → *Add Tunnel* → *Import from file* → pick `wg0.conf`.
2. Ensure the config's `AllowedIPs` includes `10.8.0.0/24`.

3. Click **Activate**.
4. From host: `curl.exe http://10.8.0.1:50000/JeMPI/countRecords`
5. From a container:
   `docker exec mediator wget -qO- http://10.8.0.1:50000/JeMPI/countRecords`

If the host reaches JeMPI but the container does not, Docker Desktop's WSL 2 network is not routing to the Windows WireGuard adapter. Simplest fix: use Option A. (Advanced: run WireGuard **inside** WSL, or add a static route in the WSL distro.)

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4. Keycloak realm

Admin Console http://localhost:8083, bootstrap admin `admin` / `admin`.

```
curl.exe -fsS http://localhost:8083/realms/openhie/.well-known/openid-configuration
```

If the JSON's `issuer` is `http://localhost:8083/realms/openhie`, the realm exists — go to §6. Otherwise:

- Realm dropdown → **Create realm** → name `openhie` → **Create**
- Or, from Git Bash: `./keycloak/setup.sh`

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5. Realm key into OpenHIM

Git Bash:

```
./keycloak/extract-realm-pubkey.sh
```

PowerShell:

```
docker compose up -d --force-recreate openhim-core
docker compose restart mediator admin-mediator patient-portal-mediator fhir-intake-
        mediator
```

> A realm rebuild rotates Keycloak's RSA key → re-run this or every token fails `JWT could not be verified`.

OpenHIM env (already in `docker-compose.yml`): `authentication_jwt_algorithms` = `RS256`, `authentication_jwt_audience` = `account`, `authentication_jwt_issuer` = `http://localhost:8083/realms/openhie`.

Transaction-port errors:

Code	Meaning
401	token missing / bad signature / issuer or audience mismatch
403	Client mapped, but lacks a channel role
404	no channel matched the URL

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6. Per-hospital client

Git Bash. One command per hospital — NHFR code + name. Creates the Organization, keypair, Keycloak client + `sub`-override mapper, and OpenHIM Client; verifies the token `sub`:

```
./keycloak/onboard-hospital.sh 3130400221 "Bharatpur Hospital"
```

Idempotent. Output gives the private-key path — hand it to the hospital, delete your copy.

System	Value
FHIR	`Organization.identifier` = `<nhfr-id>`; `Organization.id` = server UUID
Keycloak	client `<nhfr-id>`, `private_key_jwt`, token `sub` = `<nhfr-id>`
OpenHIM	Client **ID** = `<nhfr-id>`, role `np-fhir-clients`
Mediator	`X-OpenHIM-ClientID` = `<nhfr-id>` → `GET Organization?identifier=nhfr|<nhfr-id>`

Prereqs: §1, §2c, §5.

Just an Organization, no client: `./keycloak/create-org.sh <nhfr-id> "<name>"`.

> NHFR system string: deployed stack uses `http://mohp.gov.np/fhir/NamingSystem/nhfr` (NP Core's canonical is `https://nhfr.mohp.gov.np`).

> `sub` override rejected → `onboard-hospital.sh` fails at 4/5. Fallback: OpenHIM `clientID` stays the Keycloak UUID; add `FACILITY_REGISTRY_JSON={"<uuid>":"<nhfr-id>"}` to `mediator\.env` + `docker compose up -d --force-recreate mediator`.

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7. Smoke test

Git Bash (handles `$(...)` and multi-line JSON):

```
TOKEN=$(./keycloak/get-jwt-client-token.sh 3130400221)

curl -k -sS -X POST https://localhost:5000/fhir/Patient \
  -H "Authorization: Bearer $TOKEN" -H 'Content-Type: application/fhir+json' \
  -d '{"resourceType":"Patient","name":[{"family":"SHRESTHA","given":["KAMANA"]}],
       "gender":"female","birthDate":"2001-06-25",
       "telecom":[{"system":"phone","value":"9800000000"}]}'
```

`201` with a UUID `id` and `nepal-health-id` in `identifier` = the full chain works. Console → **Transactions** → the request → **orchestrations**.

Postman: `node keycloak/token-server.mjs`, then `http://localhost:4099/token/3130400221`.

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8. FHIR Intake mediator (`/ingest`)

`POST /ingest` in front of the orchestrator:

Content-Type	Body	Action
`application/json`	one raw CDRS encounter row	validate → Patient + Encounter Bundle → forward
`application/fhir+json`	resource or `transaction` / `batch` Bundle	forward unmutated
other	—	`415`

Channel `NP FHIR Intake`, `^/ingest/?$`, `POST`, route `fhir-intake-mediator:4003`, role `OPENHIM_CHANNEL_ROLE` in `fhir-intake-mediator\.env` (`np-fhir-clients` to reuse §6 clients).

Raw-row behaviour: `403` if no `X-OpenHIM-ClientID`; `GET Organization?identifier=nhfr|<nhfr-id>` → `422` if 0 or >1 match; stamps `managingOrganization` / `serviceProvider` = the resolved `Organization/<uuid>`. Inbound `X-NHFR-ID` / `organization_id` ignored.

8a. Dump script (PowerShell)

```
cd FHIR_mediator
py -m venv venv
venv\Scripts\python -m pip install -r requirements.txt
Copy-Item .env.example .env
#   KEYCLOAK_CLIENT_ID=<nhfr-id>   PRIVATE_KEY_FILE=<the .pem from §6>
#   SQLITE_PATH=<path to that hospital's cdrs_sync.db>

$env:DRY_RUN=1; venv\Scripts\python main.py     # preview
Remove-Item Env:\DRY_RUN
venv\Scripts\python main.py                     # raw rows
venv\Scripts\python main.py --FHIR              # client-mapped Bundles
venv\Scripts\python main.py --all               # also patients with no insurance number
```

Progress bookmarked per `encounter_number`.

8b. Smoke test

Git Bash:

```
TOKEN=$(./keycloak/get-jwt-client-token.sh 3130400221)

curl -k -sS -X POST https://localhost:5000/ingest \
  -H "Authorization: Bearer $TOKEN" -H 'Content-Type: application/json' \
  -d '{"hospital_patient_number":"2405078","encounter_number":"SMOKE-1",
       "first_name":"CHANDRIKA","last_name":"KANDEL","sex":"Female",
       "department_type":"C","department_id":"22","department_name":"Family Medicine",
       "visit_date_ad_time":"2026-07-28 00:02:00.000"}'
```

`201`. A second POST of the same `encounter_number` must not create a duplicate.

Quick reference

Task	Command	Shell
Create the realm	`./keycloak/setup.sh`	Git Bash
Create/update an Organization	`./keycloak/create-org.sh <nhfr-id> "<name>"`	Git Bash
Re-extract realm key	`./keycloak/extract-realm-pubkey.sh`	Git Bash

Task	Command	Shell
Onboard a hospital	`./keycloak/onboard-hospital.sh <nhfr-id> "<name>"`	Git Bash
Hospital token	`./keycloak/get-jwt-client-token.sh <nhfr-id>`	Git Bash
Postman token helper	`node keycloak/token-server.mjs` (`:4099`)	any
Recreate a mediator	`docker compose up -d --force-recreate mediator`	PowerShell
Push data	`cd FHIR_mediator; venv\Scripts\python main.py` (`--FHIR`, `--all`)	PowerShell

Troubleshooting

Symptom	Fix
`curl` returns an HTML object / weird errors	you hit `Invoke-WebRequest` — use `curl.exe`
`./keycloak/*.sh` "not recognized"	run it from **Git Bash**, not PowerShell
Console calls fail, network error	accept the cert — https://localhost:8080/heartbeat (2a)
Mediator missing under **Mediators**	`OPENHIM_PASSWORD` ≠ root password (2c)
`.env` change ignored	used `restart`, not `up -d --force-recreate`
`JWT could not be verified`	`realm-public.pem` stale — re-run §5
401 at `:5000` fresh token	issuer mismatch — `iss` must be `http://localhost:8083/realms/openhie`
403 at `:5000`	Client lacks role — re-run `onboard-hospital.sh` (§6)
404 at `:5000/ingest`	`NP FHIR Intake` channel missing — check `fhir-intake-mediator`
403 at `:5000/ingest`	set `OPENHIM_CHANNEL_ROLE=np-fhir-clients` on `fhir-intake-mediator`
422 at `:5000/ingest`	no Organization with NHFR identifier `<nhfr-id>` — `create-org.sh` (§6)
container can't reach `10.8.0.1`, host can	WSL 2 not routing to the WireGuard adapter — use §3 Option A
Keycloak realm gone after a stop	`docker compose down` was used — use `stop` / `start`